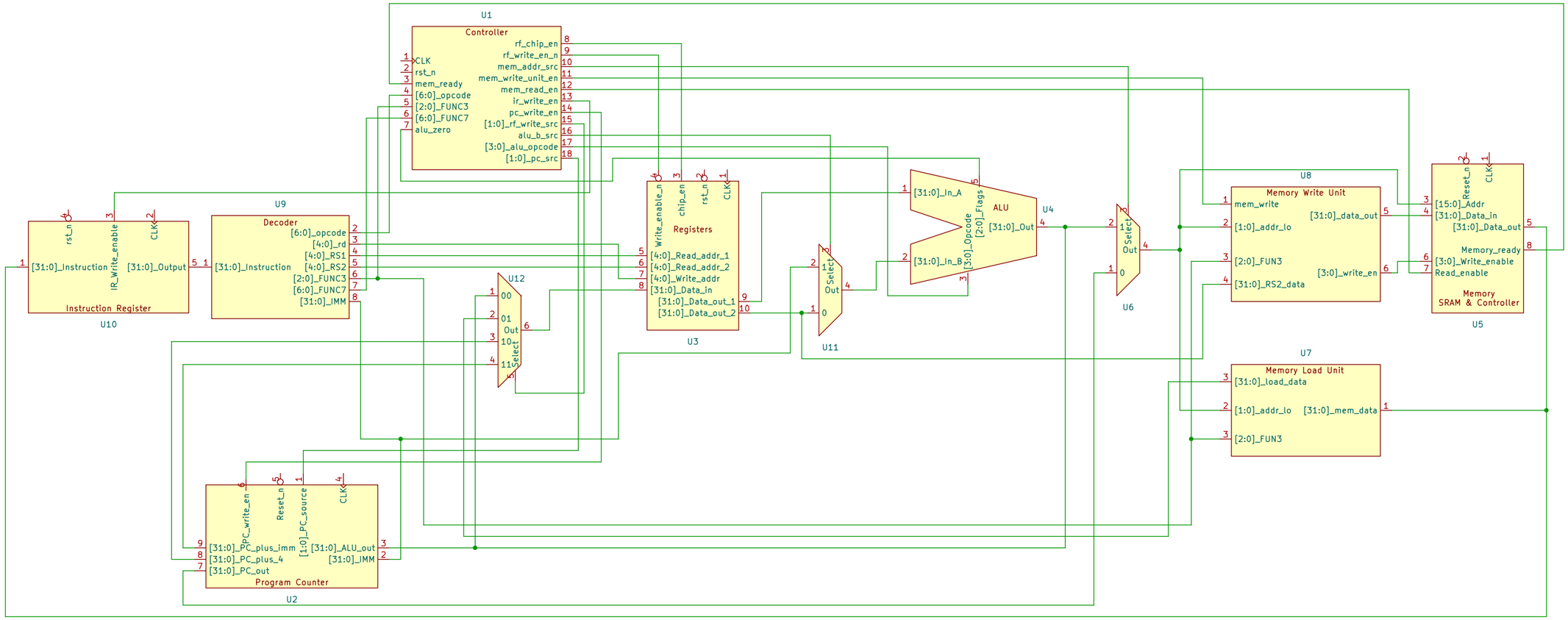
U1
Controller
CLK
rst_n
mem_ready
[6:0]_opcode
[2:0]_FUNC3
[6:0]_FUNC7
alu_zero
rf_chip_en
rf_write_en_n
mem_addr_src
mem_write_unit_en
mem_read_en
ir_write_en
pc_write_en
[1:0]_rf_write_src
alu_b_src
[3:0]_alu_opcode
[1:0]_pc_src
U9
Decoder
[6:0]_opcode
[4:0]_rd
[4:0]_RS1
[4:0]_RS2
[2:0]_FUNC3
[6:0]_FUNC7
[31:0]_IMM
[31:0]_Instruction
U10
Instruction Register
rst_n
IR_Write_enable
CLK
[31:0]_Instruction
[31:0]_Output
U12
00
01
10
11
Out
Select
U3
Registers
Write_enable_n
chip_en
rst_n
CLK
[4:0]_Read_addr_1
[4:0]_Read_addr_2
[4:0]_Write_addr
[31:0]_Data_in
[31:0]_Data_out_1
[31:0]_Data_out_2
U11
Select
Out
U4
ALU
[31:0]_In_A
[31:0]_In_B
[3:0]_Opcode
[2:0]_Flags
[31:0]_Out
U6
Select
Out
U8
Memory Write Unit
mem_write
[1:0]_addr_lo
[2:0]_FUN3
[31:0]_RS2_data
[31:0]_data_out
[3:0]_write_en
U5
Memory
SRAM & Controller
Reset_n
CLK
[15:0]_Addr
[31:0]_Data_in
[31:0]_Data_out
Memory_ready
[3:0]_Write_enable
Read_enable
U7
Memory Load Unit
[31:0]_load_data
[1:0]_addr_lo
[2:0]_FUN3
[31:0]_mem_data
U2
Program Counter
PC_write_en
Reset_n
[1:0]_PC_source
CLK
[31:0]_PC_plus_imm
[31:0]_PC_plus_4
[31:0]_PC_out
[31:0]_ALU_out
[31:0]_IMM